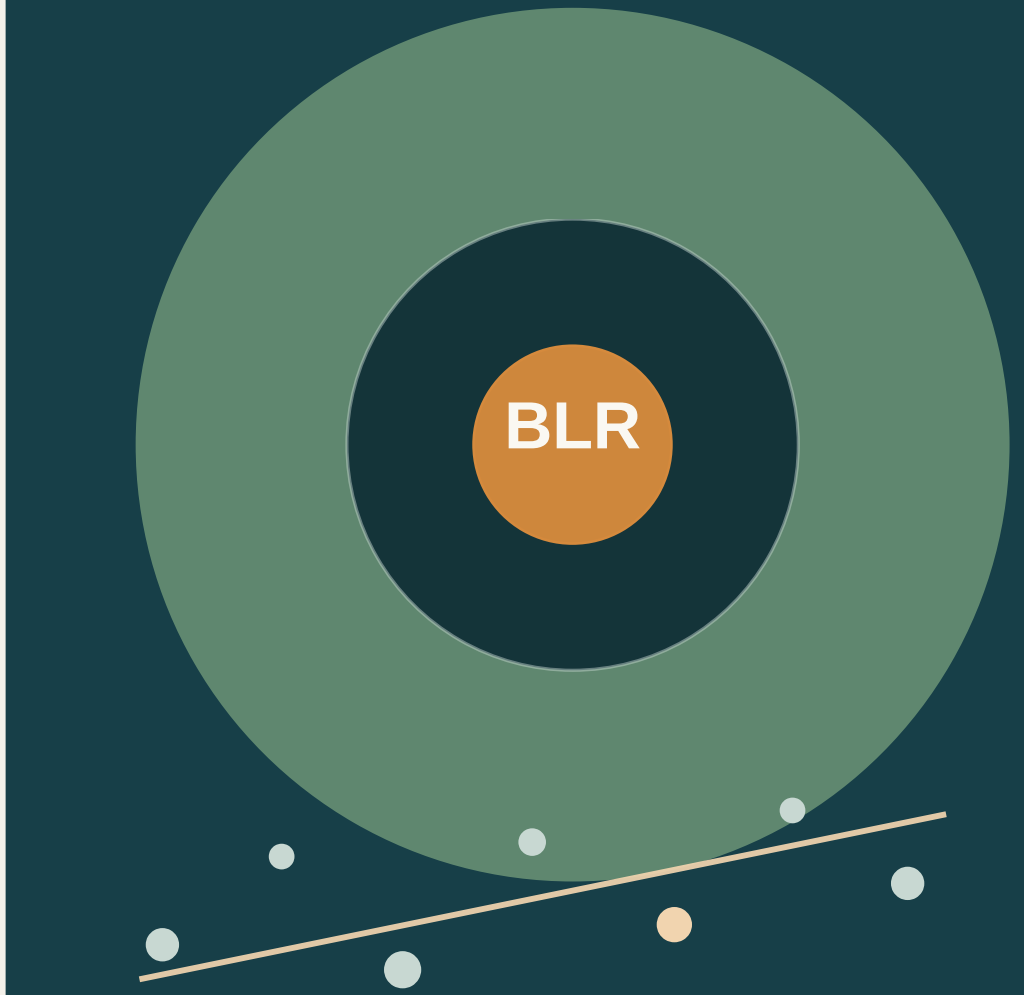
BEYOND MULTIPLE LINEAR REGRESSION

Randomized block designs

Modeling center effects and comparing treatments

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METHODS · DESIGN · STATISTICS

Why go beyond multiple linear regression?

- Experimental designs and sampling strategies in observational studies often lead to grouped observations.
- Ignoring this structure can violate the iid error assumptions, potentially biasing effect estimates or giving incorrectly estimated uncertainty.
- We use models that explicitly account for this structure, including linear mixed-effects models.
- The course develops your ability to choose, fit and interpret these models.



How the course works

A weekly cycle of preparation, guided practice and independent application.

WHEN	WHAT YOU DO
Before class	Read the selected book chapters to prepare for the session.
During class	Build understanding through lectures and develop data analysis skills in guided R labs.
After class	Apply the readings and in-class material to a new dataset in the weekly assignment.

Assignments and reproducible reporting

Use Quarto to bring your analysis and interpretation together.

ONE REPRODUCIBLE REPORT

A Quarto source file (.qmd) combines written explanations with R code.

Rendering runs the analysis and produces a report containing the results.

Explain your choices and interpret the findings alongside the analysis.

WHAT TO SUBMIT

The Quarto source file (.qmd), including all R code and annotations.

The rendered HTML report, with the R code visible.

Use the provided assignment template.

Submit both the source and the rendered report so the analysis can be read and reproduced.

Certificate and assessment

COURSE CERTIFICATE

Participate in at least 80% of the in-class meetings.

Submit the weekly data analysis assignments.

IF YOU REQUIRE A GRADE

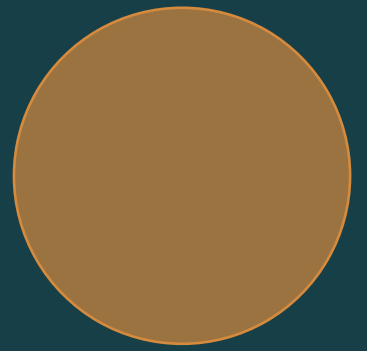
Your grade is based on the final data analysis assignment.

This applies, for example, to research master students and PhD students seeking Epidemiologist B registration.

PART 1

A multicenter trial: initial analysis





A multicenter rehabilitation trial

DESIGN AND OUTCOME

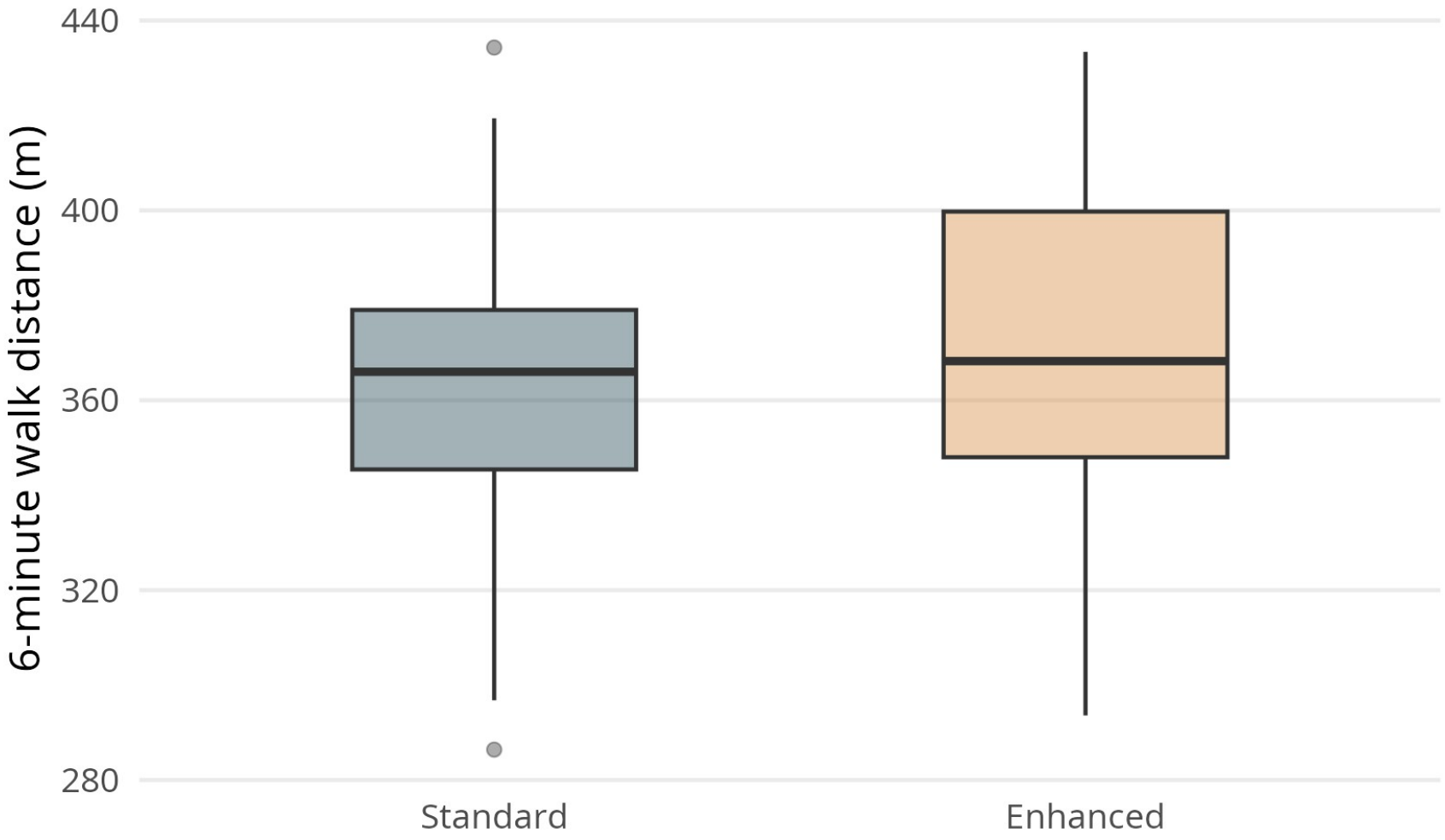
- Simulated trial: 120 patients after knee replacement at 6 centers.
- Randomized block design: centers are the blocks. Within each center, 10 patients are randomized to each treatment.
- Standard care: weekly physiotherapy and home exercises.
- Enhanced care: standard care plus a weekly supervised session.
- Outcome: six-minute walk distance at 12 weeks.

RESEARCH QUESTION

How much do the mean walking distances differ between the programs?



Walking distances pooled across centers



60 patients per program across all 6 centers.

The treatment-only model

Patient i, center j: Y_{ij} is walking distance; T_{ij} = 0 (standard) or 1 (enhanced).

SYSTEMATIC PART AND ERROR

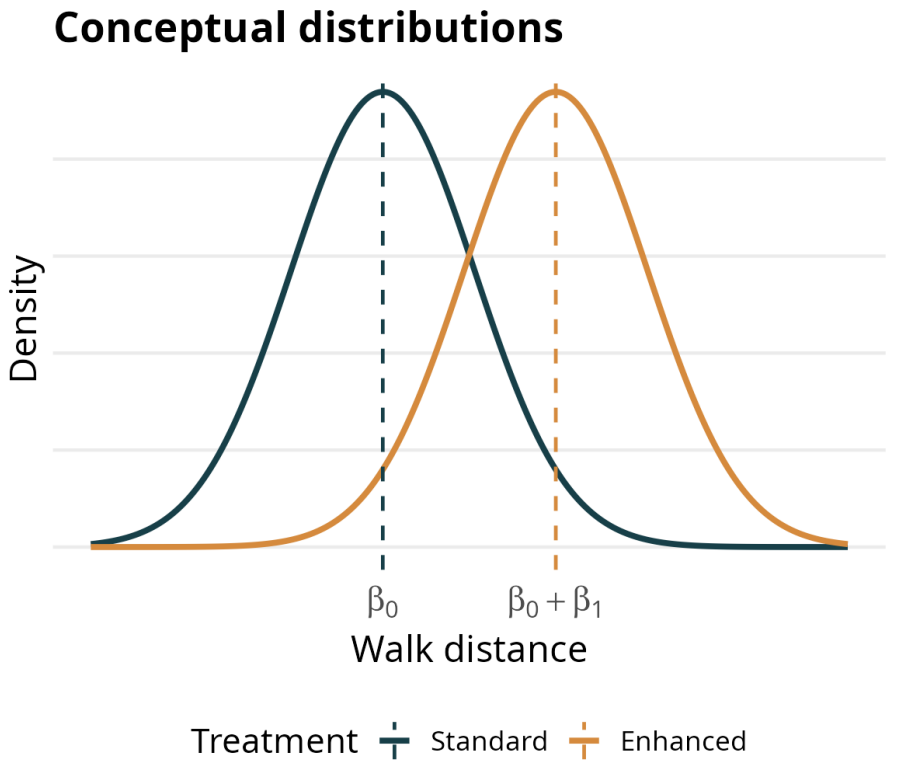
$$Y_{ij} = \underbrace{\beta_0 + \beta_1 T_{ij}}_{\text{model-predicted outcome}} + \underbrace{u_{ij}}_{\text{error}}$$

$$u_{ij} \overset{\text{iid}}{\sim} N(0, \sigma_u^2)$$

β_1 is the expected difference in walking distance between enhanced and standard care.

u_{ij} is the individual error. Errors are independent and identically distributed: Normal, with mean zero and a common variance.

MODEL-PREDICTED OUTCOMES



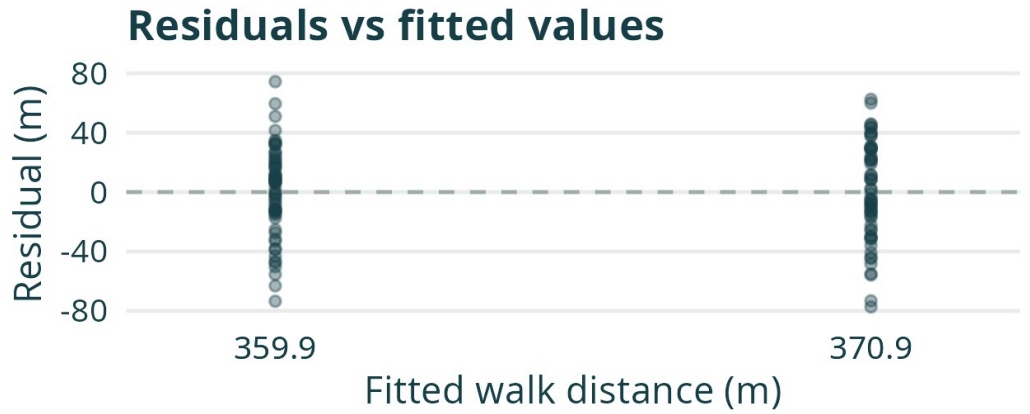
Model results

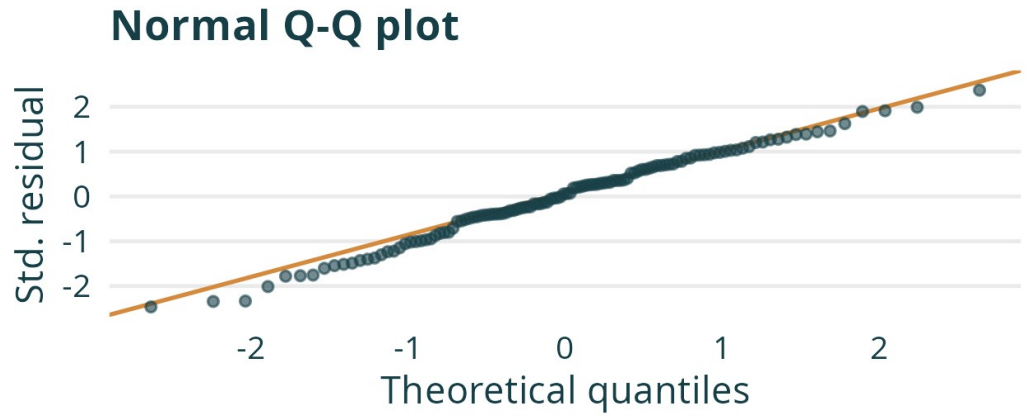
Term	Estimate (m)	SE (m)	95% CI (m)	p-value
Intercept (β_0)	359.9	4.08	351.8 to 368.0	< .001
Enhanced care (β_1)	11.0	5.77	-0.4 to 22.5	0.059

Pooling across centers, patients assigned to enhanced care walked 11.0 m farther on average.

Estimated residual variance: 1000.2 m².
Corresponding residual SD: 31.6 m.

Model diagnostics





WHAT WE USE THIS PLOT FOR

Assess whether residuals are centered around zero and have similar spread across fitted values.

WHAT WE USE THIS PLOT FOR

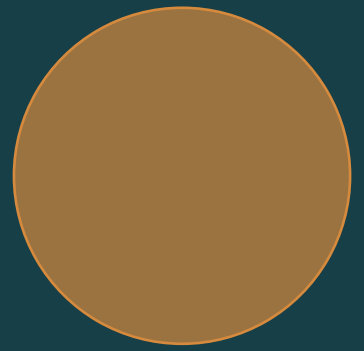
Assess whether the residual distribution is approximately Normal.

The residual plots do not suggest clear violations of the Normality or equal-variance assumptions.

PART 2

Modeling center as a fixed effect





A multicenter trial, a pooled analysis

- The trial used a randomized block design: patients were randomized to the two programs within each of six centers, with equal numbers per program in every center.
- The treatment-only model represents each treatment group as one homogeneous sample, without distinguishing the centers from which patients were recruited.
- Centers may differ in patient populations and care settings, shifting expected outcomes for both treatments.

Do center differences in expected outcomes matter for our treatment comparison?



Using simulation to investigate the consequences of ignoring center

To investigate whether ignoring center matters for the treatment comparison, we simulate repeated trials from a known model and analyse them using the treatment-only model.

STEP	SIMULATION WORKFLOW
1	Specify a data-generating model that reflects the study design.
2	Generate a dataset from that model.
3	Fit the treatment-only model, retaining the treatment estimate and its estimated SE.
4	Repeat the generation and analysis to build the estimator’s sampling distribution.

The data-generating model: fixed center effects

Patient i in center j: Y_{ij} is walking distance. T_{ij} = 0 (standard) or 1 (enhanced).

$$Y_{ij} = \beta_0 + \beta_1 T_{ij} + c_j + \varepsilon_{ij}$$

TERM	INTERPRETATION
β_0	Expected standard-care outcome averaged equally across the six centers.
β_1	Enhanced-minus-standard difference, assumed to be the same in every center.
c_j	Center j's deviation from the average standard-care outcome. The center deviations sum to zero: $c_1 + \cdots + c_6 = 0$.
ε_{ij}	Deviation from the expected outcome given treatment and center. Independent Normal errors with mean 0 and common variance σ^2.

Our experiment: new patients in the same centers

HOLD THE STUDY CONDITIONS FIXED

Six centers, 20 patients per center. Randomize 10 to each treatment within every center.

Baseline: 360 m. True treatment effect: 10 m. Independent Normal individual errors: SD 24 m.

Fixed center effects A to F (m): −17, +16, +8, −29, +27, −5.

GENERATE DATA AND FIT THE MODEL

Draw new individual errors for new patients. Keep the six center effects unchanged.

Fit the treatment-only model to every simulated study.

Retain each treatment estimate and its estimated standard error.

The generating model includes center. The treatment-only analysis omits it.

Assessing bias and calibration

BIAS: LOCATION OF THE ESTIMATES

We know the true treatment effect because we specify it in the simulation.

Compare the average treatment estimate across studies with that true effect.

Bias is the difference between the expected estimate and the true effect.

CALIBRATION OF ESTIMATED UNCERTAINTY

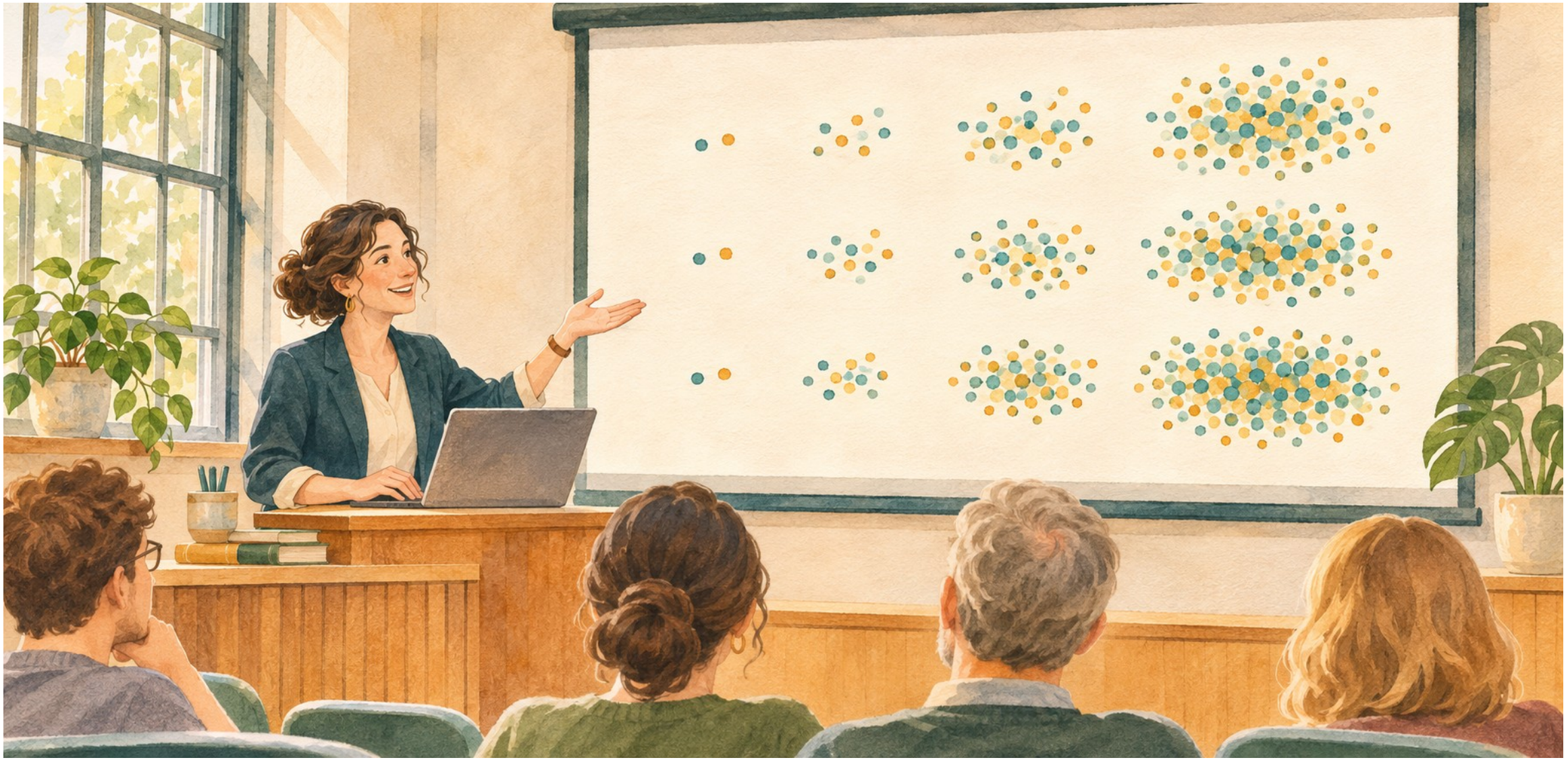
Empirical SE: the standard deviation of treatment estimates across studies.

Mean estimated SE: the average of the SEs estimated by the model across studies.

Does the estimated uncertainty match the sampling variability?

Calibration matters for confidence-interval coverage and type I error.

Live demonstration: omitting center



Summary: unbiased estimates, overstated uncertainty

ESTIMATE AND ESTIMATED SE

The treatment estimator is unbiased: estimates are centered at the true effect of 10 m.

The mean estimated SE is larger than the empirical SE.

The model overestimates the sampling variability of the treatment estimator.

CONSEQUENCES FOR INFERENCE

Consequently, its usual 95% confidence intervals overcover: they contain the true treatment effect more than 95% of the time.

When the true treatment effect is zero, its tests reject less often than the nominal 5% (type I error).

Omitting center: a mathematical explanation

TREATMENT AND CENTER

$$Y_{ij} = \beta_0 + \beta_1 T_{ij} + c_j + \varepsilon_{ij}$$

$$\varepsilon_{ij} \overset{\text{iid}}{\sim} N(0, \sigma^2)$$

Center differences are included in the expected outcome. The remaining errors satisfy the iid zero-mean Normal assumption.

TREATMENT ONLY

$$Y_{ij} = \beta_0 + \beta_1 T_{ij} + u_{ij}$$

$$\boxed{u_{ij} = c_j + \varepsilon_{ij}}$$

$$u_{ij} \sim N(c_j, \sigma^2)$$

The errors remain independent, but are no longer identically distributed: their means differ by center, while their variances remain equal.

What determines the treatment estimate’s sampling variability?

SUBSTITUTE THE DATA-GENERATING MODEL

$$\widehat{\beta}_1 = \bar{Y}_{\text{enhanced}} - \bar{Y}_{\text{standard}}$$

$$\widehat{\beta}_1 = \frac{1}{60}\sum_{j=1}^{6}\sum_{i=1}^{10}(\beta_0 + \beta_1 + c_j + \varepsilon_{ij,\text{enhanced}}) - \frac{1}{60}\sum_{j=1}^{6}\sum_{i=1}^{10}(\beta_0 + c_j + \varepsilon_{ij,\text{standard}})$$

- Both groups contain 60 copies of the same baseline, so these terms cancel.
- Both groups contain 10 copies of every center effect, so these terms cancel.

$$\widehat{\beta}_1 = \beta_1 + \bar{\varepsilon}_{\text{enhanced}} - \bar{\varepsilon}_{\text{standard}}$$

Each epsilon bar averages the individual errors used to generate that group’s outcomes.

$$\text{SE}(\widehat{\beta}_1) = \sigma\sqrt{\frac{1}{60} + \frac{1}{60}}$$

The SE estimated by the treatment-only model

THE MODEL'S CALCULATION

The treatment-only model assumes:

$$u_{ij} \overset{\text{iid}}{\sim} N(0, \sigma_u^2)$$

$$\widehat{\beta}_1 = \beta_1 + \bar{u}_{\text{enhanced}} - \bar{u}_{\text{standard}}$$

$$\text{Var}_{\text{model}}(\widehat{\beta}_1) = \frac{\sigma_u^2}{60} + \frac{\sigma_u^2}{60}$$

The estimated SE is therefore:

$$\boxed{\widehat{\text{SE}}(\widehat{\beta}_1) = \widehat{\sigma}_u \sqrt{\frac{1}{60} + \frac{1}{60}}}$$

WHAT THE RESIDUAL SD CAPTURES

- $\widehat{\sigma}_u$ is the residual SD from the treatment-only model.
- This model fits one mean for each treatment group. Differences between centers remain in the residuals.
- Its residual SD therefore reflects both center differences and individual variation.

Center differences cancel from the treatment comparison, but remain in the residual variation used to calculate its SE. The estimated SE is therefore too large.

Time for lab 1



Lab 1 debrief: accounting for center

WHAT CHANGES IN THE MODEL

- Including center allows the fitted outcomes to reflect systematic differences between centers.
- Those differences no longer remain in the residuals.
- The residual SD now estimates the individual-error SD:

$$\widehat{\sigma} = 23.8 \text{ m}$$

WHAT CHANGES IN THE SE

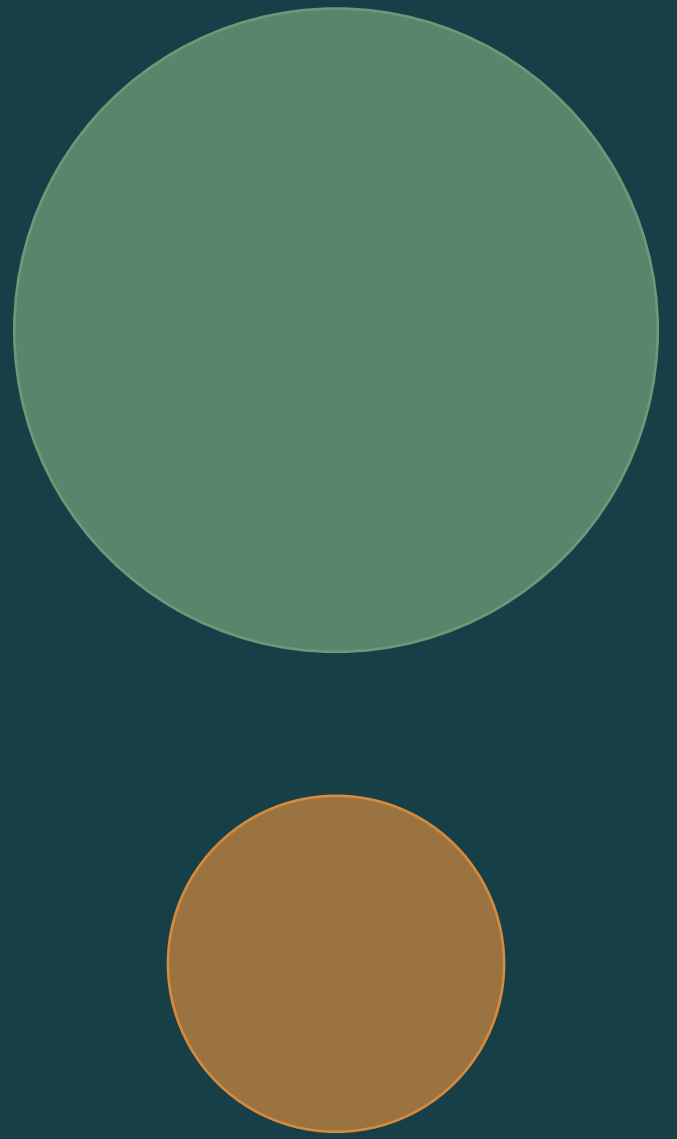
- The treatment estimate remains 11.0 m, with the same sampling distribution: balanced allocation already makes center contributions cancel.
- The SE now uses the estimated individual-error SD:

$$\widehat{\mathrm{SE}}(\widehat{\beta}_1) = \widehat{\sigma}\sqrt{\frac{1}{60} + \frac{1}{60}} = 4.34 \text{ m}$$

Including center makes the SE better calibrated: the residual variance now estimates the variation that actually contributes to the treatment comparison in this randomized block design.

PART 3

Modeling center as a random effect



From individual centers to a population of centers

- Besides estimating the treatment effect, the fixed-effects model estimates an effect for each of the six included centers.
- Often, our scientific interest is not in these particular centers or their individual effects.
- A random-intercept model accounts for center differences by describing how center effects vary across a population of centers.
- This gives a more parsimonious model and an estimate of the between-center variance.



The random-intercept model

RANDOM-INTERCEPT MODEL

$$Y_{ij} = \beta_0 + \beta_1 T_{ij} + c_j + \varepsilon_{ij}$$

$$c_j \overset{\text{iid}}{\sim} N(0, \sigma_c^2)$$

$$\varepsilon_{ij} \overset{\text{iid}}{\sim} N(0, \sigma^2)$$

For a given center effect, the expected walking distance is $\beta_0 + \beta_1 T_{ij} + c_j$. This is the **conditional mean**.

Individual errors are independent of each other and of the center effects.

IMPLIED MARGINAL MODEL

$$Y_{ij} = \beta_0 + \beta_1 T_{ij} + u_{ij}$$

$$u_{ij} = c_j + \varepsilon_{ij}$$

Averaging over center effects gives the expected walking distance $\beta_0 + \beta_1 T_{ij}$, because center effects have mean zero. This is the **marginal mean**.

Patients in the same center share c_j, so their combined errors u_{ij} are correlated.

The implied marginal model has the same mean structure as the treatment-only model, but its errors are correlated within centers.

Between-center variation and the ICC

VARIANCE COMPONENTS

$$\sigma_c^2$$

Variation between center intercepts.

$$\sigma^2$$

Individual variation within centers.

After accounting for treatment, the total variance is:

$$\mathrm{Var}(u_{ij}) = \sigma_c^2 + \sigma^2$$

INTRACLASS CORRELATION

$$\mathrm{ICC} = \frac{\sigma_c^2}{\sigma_c^2 + \sigma^2}$$

The proportion of remaining outcome variance attributable to differences between centers.

The correlation between outcomes of two patients in the same center, after accounting for treatment.

Time for lab 2



Lab 2 debrief: center variation and treatment uncertainty

INTERPRETING THE ICC

$$\widehat{\text{ICC}} = \frac{514}{514 + 565} \approx 0.48$$

About 48% of the outcome variance remaining after accounting for treatment is attributable to differences between centers.

The estimated correlation between outcomes of two different patients in the same center is 0.48, after accounting for treatment.

COMPARING THE MODELS

Model	Estimate (m)	Standard error (m)
Treatment only	11.0	5.77
Fixed center effects	11.0	4.34
Random center effects	11.0	4.34

Balanced treatment allocation within centers makes the treatment estimates identical.

Both center models separate center variation from individual variation, giving better-calibrated standard errors in this example.

The treatment results agree here, but the models describe center variation differently: separate parameters for these six centers versus a distribution for a population of centers.

Summary: Grouped data and statistical inference

GROUPED OBSERVATIONS

Grouped data arise when multiple observations come from the same unit or setting, which may contribute to their outcomes.

EXAMPLES

Patients treated at the same center.

Pupils attending the same school.

Repeated measurements from the same person.

THE SETTING WE CONSIDERED

Groups differ in their expected outcome, while the effect of the explanatory variable is the same across groups (no interaction between group and the explanatory variable).

After accounting for the explanatory variable and the group effects, the remaining errors are independent and identically distributed.

These group contributions need to be accounted for to obtain appropriate statistical inference.

Summary: Viewing group effects as fixed

HOW WE VIEW THE GROUPS	We want to account for differences between the specific groups in our study, without generalizing those differences to a wider population of groups. The individual group effects may themselves be of interest.
WHAT HAPPENS IF WE IGNORE THEM?	The errors remain independent, but have different means across groups. They are therefore no longer identically distributed. When the explanatory variable is unevenly distributed across groups, group differences can enter the estimated effect and introduce bias. Even when group contributions cancel in the comparison and the estimate remains unbiased, its usual standard error can be incorrect.
HOW WE ACCOUNT FOR THEM	Include a separate fixed effect for each group to represent its contribution to the expected outcome.

Ignoring fixed group differences can affect both the estimated effect and its estimated uncertainty.

Summary: Viewing group effects as random

HOW WE VIEW THE GROUPS	We consider the observed groups as examples from a wider population. Their contributions to the expected outcome vary randomly across groups. The standard random-intercept model assumes that these group effects have mean zero and are unrelated to the explanatory variable.
WHAT HAPPENS IF WE IGNORE THEM?	The errors have the same distribution, but are dependent within groups because observations share a random group effect. OLS remains unbiased when averaging over the random group effects, even when the explanatory variable is unevenly distributed across groups. Its usual standard error can nevertheless be incorrect because it assumes independent errors.
HOW WE ACCOUNT FOR THEM	Use a random intercept to model the variation between groups and the resulting dependence within groups.

Ignoring grouping can invalidate standard errors, confidence intervals, and tests, even when the estimated effect itself remains appropriate.

The End

Questions



